

REZA KARBASI

Data Scientist

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SUMMARY

Data Scientist with 4+ years of experience developing AI solutions across robotics, computer vision, and Speech signal processing. Master's in AI and Robotics. Proficient in **Python** and **C/C++**. Contributed to open source projects like [DeepFace](#). Adept at collaborating in multidisciplinary and international teams.

EXPERIENCE

Machine Learning Engineer

05/2023 - Now

[PresentID](#)

Remote, France

- Implemented car tracking in surveillance cameras, fine-tuned **Ultralytics** to accurately identify different lanes and vehicles.
- Enhanced an OCR system using **EasyOCR** to detect and scan official documents. Integrated **pytest** for robust testing, utilized **MLflow** for experiment tracking, and synthesized datasets for model fine-tuning.
- Automated audio file categorization by gender and age using Google's **YamNet** (in the TensorFlow framework) as the backbone, fine-tuning the model with the company's labeled data. Developed a custom metric to accurately evaluate classification accuracy, with equal emphasis on precision and recall.
- Implemented solar panel fault detection using thermal imaging, fine-tuned **YOLOv8** to segment panels and localize faults such as hotspots, diode faults, and terminal faults. Developed a tracker to pinpoint faulty panels in reports, leveraging FLIR thermal camera data from a mobile device. Successfully deployed on a 12.7 MW solar farm.

Automotive Software Engineer

05/2022 - 05/2023

[SoftwareMotion](#)

Remote, China

- Evaluated and integrated advanced path planning solutions, such as lattice-based and **Apollo's** EM planner. Also, developed obstacle avoidance algorithms with **DP** and **A*** techniques, validated through a custom Python-based simulator.
- Made a **CI/CD** pipeline in **Gitlab** with **GTest** for the Autosar-based codebase(**C/C++**), reducing on-field testing through automated unit and integration tests by 30%
- Built a **PyQT** visualization software for AEB and ACC systems, integrating BEV plots to drastically reduce debugging time. Processed MF4 and CAN signals (BLF, DBC) to expedite vehicle issue diagnosis.

AI Engineer

10/2022 - 12/2023

[Iran's National Elites Foundation](#)

Tehran, Iran

Led a team to develop a microservice-based vehicle monitoring platform, focusing on **graph-based path planning**. Enhanced navigation precision and responsiveness using **Dynamic Programming** and **A***, achieving a tenfold increase in processing speed.

Designer of a Trader Bot

04/2019 - 01/2020

[QDM](#)

Tehran, Iran

Engineered a Metatrader trading bot with a technical strategy, incorporating a **Deep Q Network** in **PyTorch**. Utilized a socket connection for data streaming and applied the algorithm on historical data spanning 2007-2019 in Metatrader. Documented a version on this [Medium blog](#) and this [Github repo](#).

Instructor
Aras Academy

06/2021 - 09/2021
Tehran, Iran

Presented **an Introduction to Reinforcement Learning** at Aras Academy's 2021 summer school, basing the syllabus on Sutton's book, "Reinforcement Learning: An Introduction."

NOTABLE PROJECTS

- Conducted a comprehensive data analysis on surgical outcomes for various medical centers and procedures. Assessed performance, compared surgery types, and provided insights to enhance medical service quality.
- Dockerized a **face recognition system** in videos using **Flask** and **deepface** library for identifying individuals by comparing faces with reference images, handling video processing, face embedding, matching, and output generation for applications like surveillance. [Project link](#).
- Led the web crawling of the [Soroush app](#) for the Big Data Course's final project, utilizing **Selenium** for data extraction and **Kafka** for transmission. Facilitated team collaboration for data analysis, with all contributions and code housed in our [GitHub repository](#).
- Led sentiment analysis of mobile phone reviews on Digikala for a Social Network course project, developing a metric to quantify device opinions. Spearheaded website crawling with **Selenium** and crafted the analysis network. Code available in [this repository](#).
- Implemented a SegNet neural network for **semantic segmentation** using **Pytorch** as part of a project in the Deep Learning course.

SKILLS

Languages English(Toefl score: 99/120), Persian: Native

Programming Python(Advanced), C(Advanced), C++(Intermediate), Matlab(Intermediate), Java(Familiar)

Frameworks Pytorch(Advanced), Tensorflow(Advanced), MLflow (Intermediate), ROS2 (Familiar), Simulink(Familiar)

Libraries Numpy, Pandas, Matplotlib, OpenCV

Databases SQL, InfluxDB, MongoDB

Misc Git, Linux, Docker, Software Testing, GTest, PyTest, Bash Scripting, REST API

EDUCATION

University of Tehran

MS in Artificial Intelligence and Robotics; GPA: 18.86/20

08/2019 - 01/2022

Tehran, Iran

K. N. Toosi University of Technology

BS in Electrical Engineering; GPA: 17.4/20

08/2014 - 01/2019

Tehran, Iran

HONORS & CERTIFICATES

Iran Open RoboCup: Ranked 5th in 2016's Small Sized League (SSL)

Snake League: Ranked 3rd in 2019's Snake League in Nasir Cup

The National University Entrance Exam for M.Sc. degree in AI: Ranked 28th(top 0.2%)

National University Entrance Exams in Mathematics: Ranked 888th(top 0.7%)

Certificate in Database Principles and SQL Server: [Link to the certificate](#)

Certificate in Django Web Framework: [Link to the certificate](#)